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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/265,046	06/02/2023	Taylor SCHREIBER	143085-5007	6603

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EXAMINER	
GOLDBERG, JEANINE ANNE	

ART UNIT	PAPER NUMBER
1682	

NOTIFICATION DATE	DELIVERY MODE
10/22/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/265,046	Applicant(s) SCHREIBER et al.	
	Examiner JEANINE A GOLDBERG	Art Unit 1682	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 7/21/26.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 174-193 is/are pending in the application.

5a) Of the above claim(s) 177-183,186-187 and 189-193 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 174-176,184-185 and 188 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED CORRESPONDENCE

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. This action is in response to the papers filed July 21, 2026. Currently, claims 174-193 are pending. Claims 177-186, 186-187, 189-193 have been withdrawn as drawn to non-elected subject matter.

Election/Restrictions

3. Applicant's election without traverse of monitoring tumor size (Claim 175) and Trim7 small molecule (Claim 185), Claims 174-176, 184-185, 188 in the paper filed July 21, 2026 is acknowledged.

The requirement is still deemed proper and is therefore made FINAL.

Priority

4. This application claims priority to

Domestic Priority data as claimed by applicant
This application is a 371 of PCT/US21/61834 12/03/2021
which claims benefit of 63/276,066 11/05/2021
and claims benefit of 63/215,735 06/28/2021
and claims benefit of 63/173,090 04/09/2021
and claims benefit of 63/121,083 12/03/2020

Drawings

5. The drawings are acceptable.

Claim Rejections - 35 USC § 112- Second Paragraph

The following is a quotation of 35 U.S.C. 112(b):

> (B) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

6. Claims 174-176, 184-185, 188 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor, or for pre-AIA the applicant regards as the invention.

A) Claim 174 is directed to treating a cancer in a subject by withdrawing Trim7 modulator. It is unclear whether absence of administering an agent is a treatment. Further, Claim 174 encompasses withdrawing Trim7 modular administration. It is unclear whether the subject had been administered Trim7 modulator and then ceased or whether the claim encompasses never administering the modulator. Claims 175-176, 184-185, 188 are similarly indefinite. Clarification is required. Applicant is reminded no new matter may be entered.

B) Claim 176 does not appear to require an additional step but merely recites an intended property of monitoring a tumor size. It is unclear what additional limitation is required to be performed or how Claim 176 further limits Claim 175. Clarification is required.

Claim Rejections - 35 USC § 102

7. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

8. Claim(s) 174-176, 184-185, 188 is/are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Webster et al. (US 2016/0303124, October 20, 2016).

Webster teaches analysis of tumor size following treatment of cancer in a subject with a treatment that inhibits function and/or activity of PD-1. Webster teaches evaluating the efficacy of a MNK-specific inhibitor in cancer (para 287). Tumors were treated and tumors were measured using a caliper (para 287). Table 3 illustrates the treatment of a subject with Cmpd 107 and anti-PD1 reduces the tumor volume/size.

With respect to Claims 184-185, 188, the claim does not require administering Trim7 modulator if there is a reduction in size of the tumor. So, no Trim7 inhibitor is administered.

TABLE 3

Activity of MSX-Specific Inhibitor Alone and in Combination with Anti-PD-1 Antibody in a Mouse Colon Carcinoma Allograft Tumor Model

Treatment	Dose	Body Weight Change[a] Average %	Tumor Growth Inhibition (% TGI)[c]	Mean Tumor Volume[c] (mm^3) ± SEM	p-value[d]
Vehicle	—/day	13.8	—, —	727.6 ± 233.5; 2598.1 ± 743.2	—
Cmpd 107	1 mg/kg/day	13.9	34, 54	522.3 ± 174.4; 1200.8 ± 489.1	0.354
Anti-PD-1	0.5 mg/mouse[b]	21.1	23, 33	586.6 ± 275.3; 1768.6 ± 991.3	0.864
Cmpd 107 + anti-PD1	1 QD/ 0.5[b]	3.8	99, 103	151.9 ± 55.0; 38.8 ± 17.5	0.0003

[a]anti-PD-1 antibody was dosed intraperitoneally once every 4 days (Q4D)
[b]Group average % body weight change calculated on final day of study (day 23)
[c]TGI and mean tumor volume was calculated 15 and 22 days after initiation of dosing
[d]p-value was calculated using a two-way ANOVA, no matching, corrected for multiple comparisons using Dunnett t-test (2-sided, equal variance assumed), relative to vehicle (day 22)

Conclusion

9. No claims allowable over the art.

10. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

A) Nguyen et al. (Abstract 6584: Aberrant Trim7 expression potentiates RACO-1 mediated proliferation and dysregulated interferon responsiveness in the setting of anti-PD-1 acquired resistance in cancer, March 22, 2024). This abstract describes applicants work post-filing.

B) Chen et al. (Trim7 Inhibition Blocks RTK/RAS pathway driven tumor cell proliferation independent of mutation and restores tumor-intrinsic IFN responsiveness, Poster #1167, November 4, 2025). This poster describes applicants work post-filing.

11. Any inquiry concerning this communication or earlier communications from the examiner should be directed to JEANINE ANNE GOLDBERG whose telephone number is (571)272-0743. The examiner can normally be reached Monday-Friday 6am-3:30pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an

interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Wu-Cheng Winston Shen can be reached on (571)272-3157. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/JEANINE A GOLDBERG/
Primary Examiner, Art Unit 1682
October 20, 2026